



|                          |                                               |
|--------------------------|-----------------------------------------------|
| • Patient's name:        | RAJIT KAUSHIK<br>RAJENDRA DEV<br>RAJENDRA DEV |
| • Received in laboratory | 20-11-2023                                    |
| • Genetic results date   | 24-11-2023                                    |

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# Fagron TrichoTest™

## Results report



## TrichoTest™ Genetic report

Fagron Genomics, S.L.U carries out genetic tests upon request by healthcare professionals, in relation to biological samples from patients obtained by the healthcare professional. Our tests do not replace a medical consultation, nor do they make up a diagnostic or treatment, nor should they be interpreted this way. Only healthcare professionals can interpret the results of said tests, based on their knowledge of the clinical records of the patients and other relevant factors and, under their responsibility, give a diagnostic or prescribe treatment to the patient. We decline all responsibility derived from the use and interpretation of the results of our tests by the solicitant healthcare professional. Fagron Genomics, S.L.U expressly reserves any legal actions in case of an innapropriate, negligent or incorrect use or interpretation of the results of our tests. It is the responsibility of the healthcare professional who requests a test to guarantee to the patient the appropriate genetic advice as foreseen by Law 14/2007, of 3rd July, of biomedical research. As Fagron Genomics, S.L.U does not have access to the personal identifiable information about the patient from whom the sample comes, it is the responsibility of the requesting healthcare professional to comply with the applicable data protection Laws and regulations.





# I. Patient identification data

# 1.

## Patient identification data



Ordering physician —●— **Dr. Sezgin Cagatay**

Contact —●— **adriana.barac@hortmanclinics.com**

Patient's name —●— **RAJIT KAUSHIK RAJENDRA DEV RAJENDRA DEV**

Gender —●— **Male**

Date of birth —●— **19-12-1982**

Sample type —●— **Buccal Swab**

Sample code —●— **BOX01039AATRI**

Received in laboratory —●— **20-11-2023**

Genetic results date —●— **24-11-2023**



## II. Recommendation of the most suitable drugs and supplements

• Patient name: **RAJIT KAUSHIK**  
**RAJENDRA DEV RAJENDRA DEV**

• Patient ID: **784198262504933**

• Date of birth: **19-12-1982**

• Sample code: **BOX01039AATRI**

• Received in laboratory: **20-11-2023**

• Genetic results date: **24-11-2023**

## 2.

### Recommendation of the most suitable drugs and supplements

The **genetic test** uses an automated qualitative pharmacogenetic algorithm that analyzes the patient's genetic data and combines this information with relevant patient history to recommend the most suitable active ingredients. Next, we show on a color scale which compounds the algorithm recommends the most. The transition from white to dark green indicates drugs from least recommended to most recommended. Medications blocked due to intolerances or contraindications are shown in red.

#### Anti-alopecic drugs

| Prostaglandins       |     |
|----------------------|-----|
| • Latanoprost Fagron | 67% |
| • Prostaquinon TM    | 50% |
| • Cetirizine Hcl     | 50% |
| • Minoxidil          | 46% |

| Antiandrogenic   |      |
|------------------|------|
| • Spironolactone | 100% |
| • Finasteride    | 99%  |
| • Saw Palmetto   | 53%  |
| • Melatonin      | 25%  |
| • Dutasteride    |      |
| • Ginseng        |      |

| Anti-inflammatory            |  |
|------------------------------|--|
| • Clobetasol propionate      |  |
| • Triamcinolone acetonide    |  |
| • Hydrocortisone             |  |
| • Betamethasone dipropionate |  |
| • Fluocinolone acetonide     |  |
| • Prednicarbate              |  |

#### Hair care supplements

| Circulation              |     |
|--------------------------|-----|
| • Arginine               | 85% |
| • L-Carnitine L-tartrate | 85% |
| • Ginkgo biloba          | 75% |
| • Caffeine               | 38% |

| Collagen synthesis   |  |
|----------------------|--|
| • Oral SiliciuMax TM |  |
| • Cystine            |  |

| Insulin-like growth factor increase |  |
|-------------------------------------|--|
| • IGrantine-F1 TM                   |  |
| • TrichoXidil                       |  |

Blocked



Recommended



Vitamin, mineral and antioxidant supplements

| Vitamin deficiency             |      | Antioxidant      |      |
|--------------------------------|------|------------------|------|
| • Vitamin B12 (Cyanocobalamin) | 100% | • Selenium yeast | 100% |
| • Vitamin E (Tocoferol)        | 67%  | • Astaxanthin    | 67%  |
| • Retinol palmitate            |      |                  |      |
| • Vitamin B9 (Folate)          |      |                  |      |
|                                |      | Keratolytic      |      |
|                                |      | • Tretinoin      |      |
| Minerals                       |      |                  |      |
| • Zinc sulfate                 | 100% |                  |      |







### III. Formulas for personalized treatment

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### 3. Formulas for personalized treatments

The pharmacogenetic algorithm has selected a series of formulations for topical, oral use or capillary mesotherapy for the care and hygiene of your patient's scalp. These personalized formulations have been selected taking into account the genetics, the type of alopecia, and the relevant history of the patient.

#### Oral treatment

| Formula                      |          |
|------------------------------|----------|
| Zinc sulfate                 | 50 mg    |
| Astaxanthin                  | 10 mg    |
| Vitamin B12 (Cyanocobalamin) | 1000 mcg |
| Selenium yeast               | 100 mg   |

| Posology                                    |  |
|---------------------------------------------|--|
| 1 capsule per day, 90 capsules for 3 months |  |

| Signature of the prescribing physician |  |
|----------------------------------------|--|
| Dr                                     |  |
| Physician registration No.             |  |
| Date                                   |  |

| Address | Signature |
|---------|-----------|
|         |           |
|         |           |
|         |           |
|         |           |

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Topical treatment

| Formula                                                                                                                |         |
|------------------------------------------------------------------------------------------------------------------------|---------|
| Latanoprost Fagron                                                                                                     | 0.005 % |
| Spironolactone                                                                                                         | 1 %     |
| Arginine                                                                                                               | 2 %     |
| TrichoSol                                                                                                              | 100ml   |
| Posology                                                                                                               |         |
| Apply at night before bedtime. Leave the solution on your scalp for as long as possible. Wash your scalp the next day. |         |

| Signature of the prescribing physician |  |
|----------------------------------------|--|
| Dr                                     |  |
| Physician registration No.             |  |
| Date                                   |  |

| Address                                         | Signature |
|-------------------------------------------------|-----------|
| <div></div> <div></div> <div></div> <div></div> |           |

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## Scalp care and hygiene

### Topical treatment

#### Formula

|                       |             |
|-----------------------|-------------|
| Ginkgo biloba         | 2.5 %       |
| Vitamin E (Tocoferol) | 3.5 %       |
| <b>TrichoOil</b>      | <b>30ml</b> |

#### Posology

1-2 times / week, massage for 3-5 minutes and leave it on for 10 min before washing your hair.

#### Signature of the prescribing physician

|                                   |  |
|-----------------------------------|--|
| <b>Dr</b>                         |  |
| <b>Physician registration No.</b> |  |
| <b>Date</b>                       |  |

#### Address

#### Signature

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Scalp care and hygiene

Topical treatment

| Formula                                         |           |
|-------------------------------------------------|-----------|
| L-Carnitine L-tartrate                          | 2 %       |
| TrichoWash                                      | 250mL     |
| Posology<br>Massage for 2 minutes and rinse     |           |
| Signature of the prescribing physician          |           |
| Dr                                              |           |
| Physician registration No.                      |           |
| Date                                            |           |
| Address                                         | Signature |
| <div></div> <div></div> <div></div> <div></div> |           |



## IV. Complete data

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## 4.

### Complete data

Data from the medical questionnaire

#### Patient demographics

Gender ☒ Male

Age (years) ☒ 42

Height (cm) ☒ 179

Weight (kg) ☒ 89

BMI ☒ 27.78

Family history of alopecia ☒ Parents

#### Hair loss data

Type of alopecia ☒ Androgenic alopecia

Grade of alopecia ☒ Grade VI

Time elapsed since start of hair loss ☒ +1

Prescription of testosterone derivatives ☒ No

#### Norwood-Hamilton Scale



Type I



Type II



Type III



Vertex



Type IV



Type V



Type VI



Type VII

#### Clinical examination

Amount of hair loss ☒ Little bit

Complaints associated with alopecia ☒ No

Patchy alopecia ☒ No

Current anti-alopecia treatment ☒ No

Previous anti-alopecia treatment ☒ No

4.

Complete data  
Pharmacogenetic results

1. Anti-alopecic drugs

Treatment efficacy with prostaglandin inhibitors

| Prostaglandin D2 |                  |                   |                  |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------|------------------|-------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gene             | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                                                                                                 |
| GPR44-1          | rs545659 (T>C)   | C                 | AA               | Genetic result: Predisposition to normal GPR44 mRNA stability. Interpretation: Prostaglandin D2 receptor 2 (GPR44 or CRTH2) variants are associated with an increased GPR44 mRNA stability leading to an increased responsiveness to prostaglandin D2 and hair follicle regression. Treatment/dosage: SNP analysis does not indicate the necessity to treat with prostaglandin D2 inhibitors.                          |
| GPR44-2          | rs533116 (C>T)   | T                 | AA               | Genetic result: Predisposition to higher GPR44 mRNA stability. Interpretation: Prostaglandin D2 receptor 2 (GPR44 or CRTH2) variants are associated with an increased GPR44 mRNA stability leading to higher responsiveness to prostaglandin D2 and hair follicle regression. Treatment/dosage: Treatment with prostaglandin D2 inhibitors (Cetirizine and/or Prostaquinon) at high doses would be highly recommended. |

| Latanoprost |                  |                   |                  |                                                                                                                                                                                                                                                                                                    |
|-------------|------------------|-------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gene        | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                             |
| PTGFR-1     | rs6686438 (T>G)  | G                 | GT               | Genetic result: Increased likelihood of having a positive response to Latanoprost. Interpretation: Prostaglandin F receptor (PTGFR) variants are related with Latanoprost treatment efficacy (prostaglandin analog) . Treatment/dosage: Treatment with Latanoprost at normal doses is recommended. |
| PTGFR-2     | rs1328441 (C>T)  | T                 | GA               | Genetic result: Increased likelihood of having a positive response to Latanoprost. Interpretation: Prostaglandin F receptor (PTGFR) variants are related with Latanoprost treatment efficacy (prostaglandin analog) . Treatment/dosage: Treatment with Latanoprost at normal doses is recommended. |
| PTGFR-3     | rs10782665 (T>G) | G                 | GT               | Genetic result: Increased likelihood of having a positive response to Latanoprost. Interpretation: Prostaglandin F receptor (PTGFR) variants are related with Latanoprost treatment efficacy (prostaglandin analog) . Treatment/dosage: Treatment with Latanoprost at normal doses is recommended. |



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### Treatment efficacy with minoxidil

| Minoxidil |                  |                   |                  |                                                                                                                                                                                                                                                                                                         |
|-----------|------------------|-------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gene      | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                  |
| PTGES2    | rs13283456 (C>T) | T                 | CC               | Genetic result: Predisposition to normal PGE2 levels.<br>Interpretation: Prostaglandin E synthase 2 (PTGES2) variants are associated with lower prostaglandin E2 production (hair growth promoter).<br>Treatment/dosage: SNP analysis does not indicate a necessity to treat with Minoxidil.            |
| SULT1A1   | rs1042028 (C>T)  | T                 | TC               | Genetic result: Predisposition to reduced SULT1A activity.<br>Interpretation: Minoxidil Sulfotransferase Enzyme (SULT1A1) variants predict response to minoxidil treatment.<br>Treatment/dosage: Minoxidil may be recommended but in high doses, with the aim of trying to ensure therapeutic activity. |

### Treatment efficacy with glucocorticoid anti-inflammatories

| Glucocorticoides |                  |                   |                  |                                                                                                                                                                                                                                                                                                                                                |
|------------------|------------------|-------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gene             | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                         |
| GR-alpha         | rs6198 (T>C)     | C                 | AA               | Genetic result: Predisposition to normal sensibility to glucocorticoid anti-inflammatory treatments.<br>Interpretation: Glucocorticoid Receptor (GR or NR3C1) variants are associated with resistance or sensitivity to corticosteroids.<br>Treatment/dosage: SNP analysis indicates that normal doses of glucocorticoids should be effective. |

### Treatment efficacy with antiandrogenics

| 17-α estradiol |                  |                   |                  |                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------|------------------|-------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gene           | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                                                             |
| CYP19A1        | rs2470152 (G>A)  | A                 | TT               | Genetic result: Predisposition to low CYP19A1 activity associated with an increased production of DHT.<br>Interpretation: Aromatase (CYP19A1) variants are associated to low conversion of testosterone in estrogens and to high conversion into DHT (hair growth inhibitor).<br>Treatment/dosage: Treatment with 17-α Estradiol (aromatase inducer) at high doses is recommended. |

| Dutasteride |                  |                   |                  |                                                                                                                                                                                                                                                                                                                                                |
|-------------|------------------|-------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gene        | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                         |
| SRD5A1      | rs39848 (T>C)    | C                 | TT               | Genetic result: Predisposition to normal SRD5A1 activity.<br>Interpretation: Steroid 5α-Reductase 1 (SRD5A1) variants are associated with increased SRD5A1 activity and increased DHT levels, thus leading to predisposition to affect hair growth.<br>Treatment/dosage: SNP analysis does not indicate a necessity to treat with dutasteride. |

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#### Finasteride

| Gene   | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                                        |
|--------|------------------|-------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SRD5A2 | rs523349 (C>G)   | G                 | CG               | Genetic result: Predisposition to increased SRD5A2 activity leading to increased levels of DHT<br>Interpretation: Steroid 5 $\alpha$ -Reductase 2 (SRD5A2) variants are associated with increased SRD5A2 activity leading to increased DHT levels and hair growth inhibition.<br>Treatment/dosage: Treatment with Finasteride at normal doses is recommended. |

## 2. Hair care supplements

### Vasodilatation and blood circulation

#### Circulation stimulators

| Gene | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                                                                                       |
|------|------------------|-------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACE  | rs4343 (A>G)     | G                 | GG               | Genetic result: Predisposition to a strongly increased Angiotensin conversion activity.<br>Interpretation: Angiotensin-converting enzyme (ACE) variants are associated with increased plasma levels of angiotensin 2, an extremely potent vasoconstrictor.<br>Treatment/dosage: High doses of circulation stimulators would be recommended, such as Minoxidil, caffeine, Ginkgo biloba, Ginseng or Arginine. |

### Collagen synthesis

#### Hair strengthening supplements

| Gene   | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                       |
|--------|------------------|-------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COL1A1 | rs1800012 (C>A)  | A                 | GG               | Genetic result: Predisposition to normal collagen stability.<br>Interpretation: Collagen, type I, alpha 1 (COL1A1) variants are associated with a normal collagen fiber.<br>Treatment/dosage: SNP analysis does not indicate the necessity to supplement with hair strengthening composites. |

### Reduction of IGF-1 levels

#### Hair strengthening supplements

| Gene  | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                      |
|-------|------------------|-------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IGF1R | rs2229765 (G>A)  | A                 | GG               | Genetic result: Predisposition to normal IGF-1 levels.<br>Interpretation: Insulin-like growth factor-I (IGF-I) variants are associated with lower plasma IGF-1 levels leading to hair loss.<br>Treatment/dosage: SNP analysis does not indicate the necessity to treat with IGF-1 inducers. |

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### 3. Vitamin, mineral and antioxidant supplements

#### Vitamins

##### Vitamin A

| Gene   | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                      |
|--------|------------------|-------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CRABP2 | rs12724719 (G>A) | A                 | GG               | Genetic result: Predisposition to normal retinoic acid intracellular transport.<br>Interpretation: Cellular retinoic acid-binding protein 2 (CRABP2) variants are associated with normal retinoic acid (vitamin A) intracellular transport.<br>Treatment/dosage: SNP analysis does not indicate the necessity to supplement with vitamin A. |

##### Vitamin B7

| Gene | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                            |
|------|------------------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BTD  | rs13078881 (G>C) | C                 | GG               | Genetic result: Predisposition to normal biotinidase activity.<br>Interpretation: Biotinidase (BTD) variants are associated with normal biotin (vitamin B7) uptake from the diet.<br>Treatment/dosage: SNP analysis does not indicate the necessity to supplement with vitamin B. |

##### Vitamin C

| Gene    | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                  |
|---------|------------------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SLC23A1 | rs33972313 (C>T) | T                 | CC               | Genetic result: Predisposition to higher vitamin C serum level.<br>Interpretation: Solute carrier family 23 member 1 (SLC23A1) variants are associated with lower serum concentration of vitamin C.<br>Treatment/dosage: SNP analysis does not indicate the necessity to supplement with vitamin C. Test for serum levels of vitamin C. |

##### Vitamin B9

| Gene  | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                             |
|-------|------------------|-------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MTHFR | rs1801133 (G>A)  | A                 | GG               | Genetic result: No predisposition to folate deficiency.<br>Interpretation: Methylene tetrahydrofolate reductase (MTHFR) variants are associated with risk of folate deficiency.<br>Treatment/dosage: SNP analysis does not indicate the necessity to supplement with folate. Test serum levels of folate prior to supplementation. |

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#### Vitamin D

| Gene | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                              |
|------|------------------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GC   | rs2282679 (T>G)  | G                 | GT               | Genetic result: Predisposition to slightly lower vitamin D serum level. Interpretation: Vitamin D-binding protein (GC or DBP) variants are associated lower vitamin D serum level. Treatment/dosage: Supplementation should be considered. Test serum levels of vitamin D prior to supplementation. |

#### Vitamin B12

| Gene | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                            |
|------|------------------|-------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FUT2 | rs602662 (A>G)   | G                 | GG               | Genetic result: Predisposition to low vitamin B12 serum level. Interpretation: Galactoside 2-alpha-L-fucosyltransferase 2 (FUT2) variants are associated lower vitamin B12 serum level. Treatment/dosage: Supplementation with vitamin B12 is highly recommended. |

#### Vitamin E

| Gene | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                   |
|------|------------------|-------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ZPR1 | rs964184 (G>C)   | C                 | CG               | Genetic result: Predisposition to slightly lower vitamin E serum levels. Interpretation: Zinc Finger Protein ZPR1 variants are associated with low alpha-tocopherol (vitamin E ) serum level. Treatment/dosage: Vitamin E supplementation should be considered. Test serum levels of vitamin E prior to supplementation. |

### Antioxidants

#### Antioxidants

| Gene | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                                                            |
|------|------------------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NQO1 | rs1800566 (G>A)  | A                 | AG               | Genetic result: Predisposition to lower NQO1 enzyme activity. Interpretation: NAD(P)H dehydrogenase [quinone] 1 (NQO1) variants are associated with lower NQO1 enzyme activity and may have less effective protection against oxidative stress. Treatment/dosage: Supplementation with antioxidants would be recommended. Test serum levels of selenium prior to supplementation. |

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## Minerals

### Magnesium

| Gene | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                  |
|------|------------------|-------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MUC1 | rs4072037 (T>C)  | C                 | CC               | Genetic result: Predisposition to lower magnesium serum level. Interpretation: Mucin 1, cell surface associated (MUC1) variants are associated with lower magnesium serum level. Treatment/dosage: Supplementation with magnesium would be highly recommended. Test serum levels of magnesium prior to supplementation. |

### Zinc sulfate

| Gene    | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                          |
|---------|------------------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SLC30A3 | rs11126936 (T>G) | G                 | GG               | Genetic result: Predisposition to lower serum zinc level. Interpretation: Solute carrier family 30 member 3 (SLC30A3) variants are associated with lower serum zinc level. Treatment/dosage: Supplementation with Zinc sulfate would be highly recommended. Test serum levels of zinc prior to supplementation. |

### Iron

| Gene    | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                                                                                                             |
|---------|------------------|-------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TMPRSS6 | rs855791 (G>A)   | A                 | CC               | Genetic result: Predisposition to normal serum levels of transferrin and iron. Interpretation: Transmembrane protease, serine 6 (TMPRSS6 or matriptase-2) variants are associated with decreased serum levels of transferrin and iron. Treatment/dosage: SNP analysis does not indicate the necessity to supplement with iron. Test serum levels of iron prior to supplementation. |

### Selenium

| Gene  | SNP (transition) | Activating allele | Patient genotype | Pharmacogenetic result                                                                                                                                                                                                                                                                              |
|-------|------------------|-------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DMGDH | rs921943 (T>C)   | C                 | CC               | Genetic result: Predisposition to lower selenium serum level. Interpretation: Dimethylglycine dehydrogenase (DMGDH) variants are associated with low selenium serum level. Treatment/dosage: Selenium supplementation should be considered. Test serum levels of selenium prior to supplementation. |



## V. Methodology

## 5. Methodology

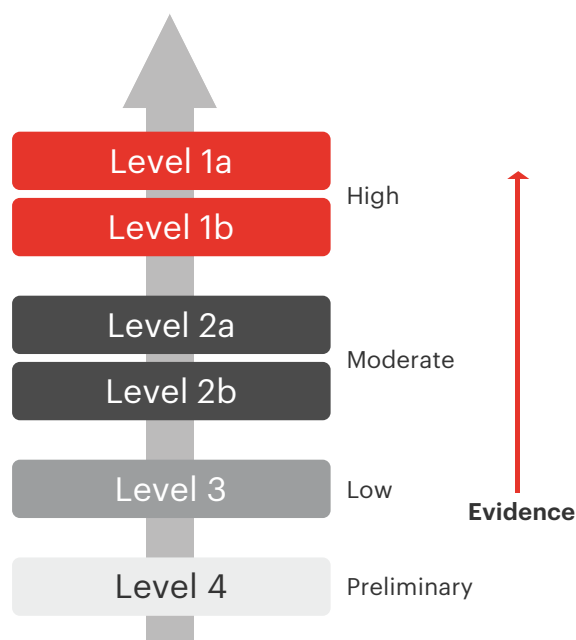
### How were the genetic variants studied selected and evaluated?

The **genetic test** was developed by a multidisciplinary team of medical doctors, pharmacists, geneticists, and programmers, following the highest quality standards. In particular, an expert team specialized in the curation of genetic variants reviewed each variant to ensure that selection, interpretation and impact of variants in the algorithms are based on the highest scientific evidence. Relevant patient's anamnesis (intolerances, diseases, medication, blood pressure, among others) that can affect recommendations was taken into account through medical questionnaires elaborated by health professionals.

- **Level 1A:** Annotation for a variant in medical society-endorsed or implemented in a major health system.
- **Level 1B:** Annotation for a variant where the preponderance of evidence shows an association. The association must be replicated in more than one cohort with significant p-values, and preferably will have a strong effect size.
- **Level 2A:** Annotation for a variant that qualifies for level 2B where the variant is within a Very Important known gene, so functional significance is more likely.
- **Level 2B:** Annotation for a variant with moderate evidence of an association. The association must be replicated but there may be some studies that do not show statistical significance, and/or the effect size may be small.
- **Level 3:** Annotation for a variant based on a single significant (not yet replicated) study or annotation for a variant evaluated in multiple studies but lacking clear evidence of an association.

- **Level 4:** Annotation based on a case report, non-significant study or in vitro, molecular or functional assay evidence only.

Only variants from level 1a to 2b were selected.



### How was this test performed?

DNA was extracted from the buccal swab sample provided and was analyzed by our clinical analysis laboratory. DNA was extracted using the KingFisher Flex® robotic extraction system (Thermo Fisher Scientific). The study of the genetic variants was carried out using a custom-designed microfluidic card to measure for the chemiluminescent detection of each of them using TaqMan® technology. TaqMan® technology for genotyping testing is proven and widely used in clinical and research settings. The sensitivity (detection limit) of this study is 99%.

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## Genetic test algorithm

The **genetic test** qualitative pharmacogenetic algorithm analyzes single nucleotide polymorphisms (SNPs) associated with metabolic pathways involved in alopecia predisposition and treatment and combines this data with relevant patient history to predict treatment responses and recommends the most appropriate active ingredients.

The **genetic test** is an in vitro diagnostic medical device developed by **Fagron Genomics** and marketed under the CE-IVD mark in conformity with European Directive 98/79/EC and the transitional provisions (article 130) of European Regulation 2017/746.



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## What are the limits of this report?

Each genetic marker tested is just one factor that predicts the likelihood of a particular outcome. However, the lifestyle, diet, and environment to which the patient is exposed may impact the expected outcomes. These external factors cannot be taken into account in this report.

The information in this report is not used to diagnose genetic diseases or abnormalities, as it does not predict the risk and likelihood of certain genetic outcomes. It is also not intended to diagnose or cure any disease. The **genetic test** is intended to assist health professionals in making patient-specific care decisions regarding the treatment or prevention of androgenetic alopecia, areata alopecia, and telogen effluvium.

Our clinical laboratory has standard and effective procedures to protect against technical and operational problems. However, problems may occur in the shipment to the laboratory or in the handling of the sample, including, but not limited to, damage to the sample, mislabeling, and loss or delay in receiving the sample. In such cases, the medical laboratory may need to request a new sample.

As with all medical laboratory tests, there is a small chance that the laboratory may provide inaccurate information.

It is the responsibility of the professional who requests a test from us to guarantee the interested party appropriate genetic counseling in accordance with Law 14/2007, of July 3, on Biomedical Research.

**Fagron Genomics S.L.U.** declines all responsibility derived from the use and interpretation of the results of our tests by the requesting health professional.

**Fagron Genomics S.L.U.** does not access data identifying the patient from whom the sample comes, so it is also the responsibility of the requesting professional to comply with the applicable data protection regulations.





## VI. References

• Patient name: **RAJIT KAUSHIK**  
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